

THE RELATIONSHIP BETWEEN EMOTIONS IN SONG LYRICS AND GENRE & POPULARITY

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Özetçe —Müzik, tarih boyunca insanların duygularını ifade etmesinde en etkili yollardan biri olmuştur. Müzikte bulunan melodi ve ritim kadar şarkı sözleri de bu duygusal ifadenin önemli bir parçasıdır. Bu çalışmada şarkı sözlerinde ifade edilen duyguların müzik türleri ve popülerlik ile ilişkisi incelenmiştir. Araştırmanın temel motivasyonu, müziğin duygusal etkisinin dinleyici ve endüstriyel başarı üzerindeki rolünü sayısal verilerle ortaya koymaktır.

Çalışma kapsamında, Paul Ekman'ın altı temel duygu modelinde yer alan öfke, iğrenme, korku, neşe, üzüntü ve şaşkınlık duyguları temel alınarak bir sınıflandırma sistemi geliştirilmiştir. Bu sistemi oluşturmak için belirtilen duygularla etiketlenmiş bir veri seti veri ön işleme adımlarından geçirilmiş, ardından sınıf dengesizliğini gidermek için veri artırımı ve veri sınırlama gibi stratejiler kullanılarak çeşitli varyasyonları elde edilmiştir. Bu veri setleri ile BERT, RoBERTa ve DeBERTaV3 modelleri eğitilip performansları karşılaştırılmıştır. Karşılaştırma sonunda en başarılı model %84 doğruluk ve %85 makro F1 skoru ile, kısmi veri artırımı ve sınırlamayla oluşturulan veri setiyle eğitilen RoBERTa modeli olmuştur. Spotify API ve Genius API ile oluşturulan gerçek dünya verisi, bu modelle etiketlenerek istatistiksel analizlere tabi tutulmuştur. Bu analizler sonucunda oluşturulan veri setindeki şarkılarda korku, üzüntü gibi olumsuz duyguların baskın olduğu, bu duyguların popülerlik ile pozitif ilişkili olduğu ve özellikle R&B ile Blues türlerinde duygusal yoğunluk ve popülerliğin yüksek olduğu gözlemlenmiştir.

Bu çalışma ile şarkı sözlerindeki ifadelerin duygusal analizi yapılmış, duygu-tür-popülerlik üçlüsü multidisipliner bir yaklaşımla ele alınmıştır. Çalışma sonucunda elde edilen bulgular, müzik endüstrisi ve akademik çalışmalar için yol gösterici nitelik taşımaktadır.

Anahtar Kelimeler—BERT, DeBERTaV3, Derin Öğrenme, Doğal Dil İşleme, Duygu Analizi, Müzik Türü, Popülerlik, RoBERTa, Şarkı Sözü

Abstract—Music has been one of the most effective ways for humans to express their emotions throughout history. In addition to melody and rhythm, lyrics are also a significant component of this emotional expression in music. In this study, the relationship between the emotions expressed in song lyrics and musical genres as well as popularity is examined. The primary motivation of the research is to reveal the role of music's emotional impact on listeners and industrial success using quantitative data.

Within the scope of the study, a classification system was developed based on Paul Ekman's six basic emotion model, which includes anger, disgust, fear, joy, sadness, and surprise. To build this system, a dataset labeled with these emotions was subjected to data preprocessing steps, and then various variations were created by applying strategies such as data

augmentation and data limitation to address class imbalance. BERT, RoBERTa, and DeBERTaV3 models were trained using these datasets, and their performances were compared. As a result of the comparison, the most successful model was the RoBERTa model trained with a dataset created through partial augmentation and limitation, achieving 84% accuracy and 85% macro F1 score. Real-world data gathered through the Spotify API and Genius API was labeled using this model and subjected to statistical analyses. The analyses showed that negative emotions such as fear and sadness were dominant in the songs in the dataset, that these emotions were positively correlated with popularity, and that emotional intensity and popularity were especially high in the R&B and Blues genres.

Through this study, an emotional analysis of the expressions in song lyrics was conducted, and the emotion-genre-popularity triad was addressed using a multidisciplinary approach. The findings obtained from the study are of guiding value for both the music industry and academic research.

Keywords—BERT, DeBERTaV3, Deep Learning, Natural Language Processing, Sentiment Analysis, Music Genre, Popularity, RoBERTa, Song Lyrics

I. INTRODUCTION

Music has been one of the most powerful tools people have used throughout history to express, share, and interpret their emotions. In vocal music, emotions are conveyed not only through melody and rhythm, but also through lyrics. Examining how emotions are expressed through lyrics is of great importance in understanding the effect of music on humans. In recent years, studies examining the relationship between music and emotion have increased, and in this context, new approaches to analyzing lyrics using natural language processing techniques have been developed.

In this project, the aim is to identify and analyze the emotions present in song lyrics. The data obtained from emotion analysis will be correlated with factors such as music genres and popularity to enable a comprehensive evaluation. The project aims to contribute to the existing literature in this field by examining the relationship between music and emotion using natural language processing methods.

II. LITERATURE REVIEW

In most studies aimed at understanding the relationship between music and emotion, the focus has been on classifying musical pieces as either positive or negative. However, in the study conducted by Edmons et al., it was

demonstrated that song lyrics can be labeled based on the eight basic emotions in Plutchik’s wheel of emotions [1]. In the study, methods such as Naive Bayes, Random Forest, and BERT were used, and it was noted that the performance of the BERT model improved significantly, particularly with a small song dataset [2].

In the study conducted by Agrawal et al., the effect of song lyrics on mood was investigated, and a transformer-based method using XLNet was applied. In the research, which utilized the MoodLyrics [3], MER, and AllMusic datasets, results of 76.31% accuracy and 75.40% F1 score were obtained for single-label classification, particularly with the AllMusic dataset. Although the multi-label classification structure enabled faster prediction by the model, the results of the single-label classification structure were more successful. The study contributes to the development of mood-based playlists and the enhancement of accuracy in music recommendation systems by utilizing the emotional characteristics of song lyrics [4].

In the study conducted by Agatha et al., it was determined that one of the factors influencing a song’s popularity is the emotions conveyed in its lyrics, and based on this finding, a model capable of emotion prediction was developed. The BERT model was fine-tuned using the SpotGenTrack [5] dataset. In the study, which involved 52,278 songs from the dataset, results of 88% precision, 87% recall, accuracy, and F1 score were achieved. These results were found to be higher compared to a previous study in which a BERT model was used and an accuracy score of 73% was obtained [6].

III. DATASETS AND PREPROCESSING

In this study, two main datasets were utilized: one for training the emotion classification model and another for conducting emotion analysis on a collection of songs. The steps applied to preprocess these datasets and their structural organization are presented in detail in this section.

A. Dataset Used for Model Training

For training the model, a publicly available emotion-labeled lyrics dataset accessed via the HuggingFace platform was used [7]. The dataset initially contained 36,897 samples. However, several preprocessing steps were applied before use. First, rows with missing values were removed, followed by the elimination of duplicate or highly similar entries. Subsequently, various data preparation procedures aimed at improving model performance were applied. The dataset originally included a ‘neutral’ label, but since the project is based on Paul Ekman’s six basic emotions, this label was removed.

To enhance class balance, the Parrot Paraphraser model [8] was employed for augmentation without altering the sentiment labels. Augmentation was applied cautiously, ensuring that no class exceeded its original sample count. In addition to increasing the size of minority classes, random downsampling was also performed on dominant classes. To improve model performance, augmentation was applied to the ‘disgust’ class due to its low sample count, resulting in a new version referred to as Dataset 1. A second version,

Dataset 2, was created by limiting the number of samples in dominant classes to 4,000 and augmenting the remaining classes. Lastly, a fully balanced dataset, Dataset 3, was formed by adjusting each class to contain 2,000 samples through both augmentation and downsampling. The sample counts for each version are given in Table 1.

Table 1 Sample Counts per Dataset for Each Emotion Class

Emotion	Initial	Preprocess	Dataset 1	Dataset 2	Dataset 3
Anger	4,023	3,342	3,341	4,045	2,000
Disgust	831	699	1,383	1,913	2,000
Fear	10,937	8,324	8,307	4,000	2,000
Joy	2,881	1,756	1,743	3,643	2,000
Neutral	5,719	0	0	0	0
Sadness	10,798	7,417	7,399	4,000	2,000
Surprise	1,706	1,306	1,303	2,990	2,000

B. Dataset Used for Emotion Analysis

After training the emotion prediction model, a second dataset comprising 7,219 songs was collected via the Spotify and Genius APIs. Lyrics were retrieved from Genius, while metadata such as title, artist, genre, popularity, and release year were obtained from Spotify. Filtering criteria were applied to ensure the inclusion of English-language songs released in the United States across six genres: pop, rock, hip-hop, R&B, country, and blues. The distribution by genre is provided in Table 2.

Table 2 Number of Samples per Genre in the Emotion Analysis Dataset

Genre	Sample Count
Pop	1,537
Blues	1,528
Country	1,393
Rock	1,269
Hip-hop	1,353
R&B	139

C. Data Preprocessing

Several preprocessing steps were applied to both the training and analysis datasets to improve data quality, consistency, and suitability for modeling. Unnecessary columns, such as technical indexing fields, were removed to simplify the dataset structure. Records with missing values were excluded to avoid training errors and misinterpretations. HTML tags, structural annotations, and special characters that did not contribute to sentiment were eliminated from the lyrics. After cleaning, excessive whitespace was normalized to ensure uniform formatting. Lastly, duplicate records—either identical lyrics or repeating combinations of artist, song title, and lyrics—were detected and removed to prevent bias in model learning.

IV. METHODS AND RESULTS

This section presents the training configurations and performance analyses of three transformer-based models—BERT, RoBERTa, and DeBERTaV3—trained on multiple balanced and augmented datasets. Following comparative evaluations, the best-performing model is selected for further power and ground truth analysis, and finally applied to a real-world dataset for emotion-based music analysis.

A. Training Details

All models were trained on three datasets constructed through different augmentation and balancing strategies. Each dataset was split into 80% training, 10% validation, and 10% test sets, ensuring class balance within each split. To mitigate class imbalance in Dataset 1 and Dataset 2, class weights were incorporated during training. Early stopping was employed based on the weighted F1 score; training was halted if the score decreased for two consecutive evaluation steps. Hyperparameters were selected based on empirical tuning and literature guidelines for each model.

B. Comparative Performance Analysis

Table 3 summarizes the macro precision, macro recall, accuracy and macro F1 score of all models across the three datasets. RoBERTa consistently outperformed the other models, achieving the highest scores on all datasets, with its best performance on Dataset 2 (accuracy = 0.84, macro F1 = 0.85).

BERT demonstrated stable and competitive performance, particularly on Dataset 2. DeBERTaV3, despite its theoretical advantages, underperformed in comparison, particularly on Dataset 3, where it showed significant performance degradation.

Table 3 Summary of Model Performances

Model	Dataset	Precision	Recall	Accuracy	F1 Score
BERT	Dataset 1	0.76	0.76	0.78	0.76
BERT	Dataset 2	0.83	0.83	0.82	0.83
BERT	Dataset 3	0.78	0.78	0.78	0.78
RoBERTa	Dataset 1	0.80	0.80	0.82	0.80
RoBERTa	Dataset 2	0.84	0.85	0.84	0.85
RoBERTa	Dataset 3	0.82	0.82	0.82	0.82
DeBERTaV3	Dataset 1	0.73	0.78	0.77	0.75
DeBERTaV3	Dataset 2	0.77	0.79	0.77	0.77
DeBERTaV3	Dataset 3	0.72	0.72	0.72	0.71

C. Impact of Dataset Balancing

Each dataset exhibited distinct characteristics influencing model performance. Dataset 1 was imbalanced, with dominant fear and sadness classes and limited augmentation on the disgust class. Dataset 2 was moderately balanced via augmentation and downsampling, leading to noticeable performance gains. Dataset 3 was fully balanced with synthetic samples (2,000 per class), but models exhibited

slightly reduced performance, suggesting that synthetic balancing may not always yield optimal generalization.

RoBERTa’s robustness across datasets, including fully balanced synthetic data, indicates strong generalization capabilities. Conversely, DeBERTaV3 was more sensitive to data quality and augmentation, showing decreased reliability under synthetic conditions.

D. Emotion-Specific Observations

The joy and sadness classes consistently achieved high F1 scores across all models, indicating clearer linguistic patterns or stronger data representation. In contrast, disgust and surprise, initially underrepresented, benefitted substantially from augmentation, particularly with RoBERTa. Performance on fear and anger varied depending on the dataset and model, with Dataset 3 posing more difficulty.

E. Power Analysis and Ground Truth

Based on the best-performing model (RoBERTa with Dataset 2), a power analysis was conducted using the NormalIndPower library with effect size = 0.8, $\alpha = 0.05$, and power = 0.8, estimating a sample requirement of 150. A random subset of 114 samples was annotated independently by team members, and a consensus was reached through majority voting, yielding a Cohen’s Kappa of 0.64. This dataset served as the ground truth reference. The model achieved a weighted F1 score of 73% on the ground truth set, indicating a satisfactory level of generalization performance.

F. Real-World Emotion Analysis

An analysis was conducted on a real-world dataset to examine the distribution of predicted emotions and their potential relationship with musical genres and popularity scores. The emotion labels assigned to the dataset revealed a dominance of negative emotions such as fear, sadness, and disgust, with fear emerging as the most frequently detected emotion. In contrast, positive emotions such as joy and surprise appeared less frequently, both in terms of assigned labels and associated average popularity scores. These findings suggest that song lyrics expressing negative emotions may be more prevalent and possibly more engaging for listeners.

When examining the average popularity scores across emotions, it was observed that songs associated with negative emotions generally received higher scores than those labeled with positive emotions. This trend indicates a listener preference for emotionally intense content with negative sentiment.

Among musical genres, R&B and Blues stood out with relatively higher popularity scores and a stronger representation of emotional content. In particular, R&B songs labeled with anger, disgust, and fear exhibited average popularity scores exceeding 70. This suggests that genres characterized by emotional depth and negative sentiment may attract more listener attention. In comparison, other genres showed lower average popularity scores and a reduced presence of positive emotional labels.

The detailed emotion counts per genre are presented in Table 4, while the corresponding average popularity scores are shown in Table 5.

Overall, the analysis demonstrates a clear influence of emotion—especially negative sentiment—on music popularity. The results indicate that emotional tagging of lyrics, along with genre characteristics, can significantly impact a song’s reception and popularity among listeners.

Table 4 Distribution of Predicted Emotions Across Music Genres

Genre	Anger	Disgust	Fear	Joy	Sadness	Surprise
Blues	269	138	616	96	226	183
Country	222	112	567	89	246	157
Hip-hop	266	105	536	99	206	141
Pop	220	82	645	111	277	202
R&B	23	13	53	9	27	14
Rock	188	69	527	78	227	180

Table 5 Average Popularity Scores (%) by Genre and Emotion Labels

Genre	Anger	Disgust	Fear	Joy	Sadness	Surprise
Blues	67.74	68.14	67.67	67.07	65.78	67.39
Country	61.23	63.39	59.43	58.73	56.47	60.11
Hip-hop	51.26	48.34	59.28	56.83	59.00	56.13
Pop	50.19	51.66	46.27	41.95	48.77	50.51
R&B	74.09	72.31	71.83	71.11	67.52	73.29
Rock	57.22	61.13	46.27	46.59	48.08	36.73

V. CONCLUSION

In this study, emotions expressed in song lyrics were analyzed to examine their relationships with musical genres and popularity. To perform emotion classification, three transformer-based models—BERT, RoBERTa, and DeBERTaV3—were fine-tuned using a pre-labeled lyrics dataset. In order to improve model performance, data augmentation and class balancing strategies were applied to create three distinct training datasets. Among the models, RoBERTa achieved the best performance and was subsequently used to annotate a broader real-world dataset collected via Spotify and Genius APIs, which included lyrics, genres, and popularity scores.

The evaluation of the results demonstrated that the RoBERTa model consistently outperformed BERT and DeBERTaV3 across all datasets, achieving its highest accuracy, precision, recall, and F1 scores—84%, 84%, 85%, and 85%, respectively—on the partially balanced Dataset 2. Compared to previous works in the literature, these results were competitive or superior. It was also observed that the data augmentation and balancing strategies applied in Dataset 2 significantly contributed to improved classification performance, especially for underrepresented emotion classes.

The emotion distribution in the real-world dataset labeled by the best-performing model revealed a dominance

of negative emotions such as fear, sadness, anger, and disgust. These emotions were also positively correlated with popularity scores, while positive emotions like joy and surprise appeared less frequently and had lower average popularity. This suggests that negative emotions may resonate more deeply with listeners and are more prevalent in popular songs.

Overall, the findings confirm that model selection and dataset design significantly impact emotion classification performance. Furthermore, emotional content and genre preferences in lyrics play a critical role in shaping how songs are received and appreciated by listeners.

VI. FUTURE WORK

The results of this study offer valuable insights for emotion-based music analysis and also highlight directions for future research. One major limitation observed was the imbalance in emotion classes, with some emotions being underrepresented. Future work should focus on developing larger and more diverse datasets with balanced emotion categories to improve classification reliability. Although strong results were achieved using BERT-based models, future studies may consider experimenting with other transformer architectures or even traditional machine learning models for comparative purposes. In addition, integrating musical features alongside lyrical content could further enhance emotion detection by combining textual and audio signals.

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